

Question 1:

Find $\int 4(x-15)^6 dx$

Hint: Let $u = x - 15$

a) $du =$ _____

b) Answer: $\int 4(x-15)^6 dx =$ _____

Question 2:

Find $\int \frac{17}{(x-1)^7} dx$

Hint: Let $u = x - 1$

a) $du =$ _____

b) Answer: $\int \frac{17}{(x-1)^7} dx =$ _____

Question 4:

$$\text{Find } \int \frac{x^2 - 5}{-x^3 + 15x + 1} dx = \int \frac{1}{-x^3 + 15x + 1} (x^2 - 5) dx$$

$$\text{Hint: Let } u = -x^3 + 15x + 1 \quad \frac{du}{dt} = -3x^2 + 15 = -3(x^2 - 5)$$

$$\text{a) } du = \underline{\hspace{2cm}}$$

$$\text{b) } \int \frac{x^2 - 5}{-x^3 + 15x + 1} dx = \underline{\hspace{2cm}}$$

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$$\text{b) } \int \frac{x^2 - 5}{-x^3 + 15x + 1} dx = \underline{\hspace{2cm}}$$

Question 5:

Find $\int \frac{4x^2}{x-1} dx$ Hint: Divide $\frac{4x^2}{x-1}$ first.

a) Divide $\frac{4x^2}{x-1} = ?$

b) Answer: $\int \frac{4x^2}{x-1} dx =$ _____

Question 6:

Find $\int \frac{e^x}{10+e^x} dx = \int \frac{1}{10+e^x} (e^x) dx$ Hint: Let $u = 10 + e^x$

a) $du =$ _____

b) Answer: $\int \frac{e^x}{10+e^x} dx =$ _____

Question 7:

Find $\int x \cdot \sin(4\pi x^2) dx$

Hint: Let $u = 4\pi x^2$

a) $du =$ _____

b) $\int x \cdot \sin(4\pi x^2) dx =$ _____

Question 8:

Find $\int \frac{\cos x}{\sqrt{\sin x}} dx = \int \frac{1}{\sqrt{\sin x}} (\cos x) dx$

Hint: Let $u = \sin x$

a) $du =$ _____

b) $\int \frac{\cos x}{\sqrt{\sin x}} dx =$ _____

Question 9:

$$\text{Find } \int \frac{\ln x^2}{x} dx = \int \ln x^2 \cdot \left(\frac{1}{x}\right) dx$$

Hint: Let $u = \ln x^2 = 2 \ln x$

a) $du =$ _____

b) $\int \frac{\ln x^2}{x} dx =$ _____

Question 10:

$$\int \frac{8x}{\sqrt{x^2 + 16}} dx = \int \frac{1}{\sqrt{x^2 + 16}} (8x) dx$$

Hint: Let $u = x^2 + 16$

a) $du =$ _____

b) $\int \frac{8x}{\sqrt{x^2 + 16}} dx =$ _____

Question 11:

$$\int \frac{1}{13+4x^2} dx$$

Hint: Use Formula $\int \frac{1}{a^2 + u^2} du = \frac{1}{a} \arctan \frac{u}{a} + C$

a) $a^2 =$ _____

b) $u^2 =$ _____

c) $\int \frac{1}{13+4x^2} dx =$ _____

Question 12:

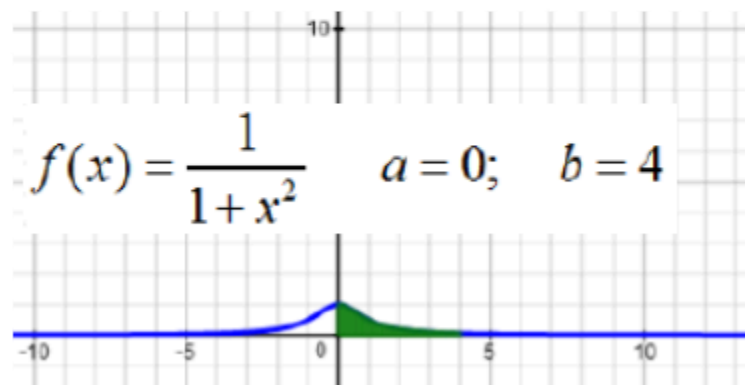
12) Find the area.

Hint: Use Formula $\int \frac{1}{a^2 + u^2} du = \frac{1}{a} \arctan \frac{u}{a} + C$

$$f(x) = \frac{1}{1+x^2} \quad a=0; \quad b=4$$

a) Set up definite integral: Area = $\int_?^? ? \, dx$

b) Evaluate integral: Area = _____



Question 13:

13) Find the area.

$$f(x) = \frac{1}{\sqrt{x^2 + 0.4}} \quad a = 0; \quad b = 2$$

a) Set up definite integral: Area = $\int_{?}^{?} ? \, dx$

b) Evaluate integral: Area = _____

